

Predicting 30-day Diabetic Hospital Readmission: A Replication Study

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Abstract

1. This replication study builds on Emi-Johnson and Nkrumah (2025) to predict 30-day diabetic hospital readmission.
2. Research question: **To what extent can electronic health record (EHR) data, specifically demographics, admission details, and medication regimens, be used to predict early (<30-day) hospital readmission for diabetic patients using supervised machine learning algorithms?**
3. Using the UCI Diabetes dataset (N = 101,766 after cleaning), XGBoost achieved a superior 0.680 AUC-ROC and 0.897 recall, which is close to the original study.
4. SHAP analysis identified prior discharge destination, inpatient visits and medication burden as primary predictors.
5. High recall but lower accuracy and precision in my replication, and lower recall but higher accuracy and precision in the original study, highlight a critical precision-recall trade-off that remains a barrier to clinical adoption of such models.



Introduction

- Diabetes remains one of the most significant global health challenges, characterized by its chronic nature and the high complexity of clinical management. Within the inpatient setting, diabetic patients often face constant readmissions.
- 30-day hospital readmission serves as a critical benchmark for patient safety and inpatient quality of care in diabetes management.
- This study is **a replication of the research conducted by Emi-Johnson and Nkrumah (2025)** using supervised machine learning.
- The objective is to determine **how accurately EHR data (demographics, admission details, and medications) can predict early readmission (< 30 days)**.
- Analysis is conducted on the UCI Diabetes 130-US Hospitals dataset, encompassing 101,766 encounters across 130 facilities.

Predicting 30-Day Hospital Readmission in Patients With Diabetes Using Machine Learning on Electronic Health Record Data

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Abstract

Introduction: Hospital readmissions within 30 days remain a critical issue in healthcare, signaling potential care discontinuities and contributing to escalating costs. Leveraging machine learning (ML) on electronic health records (EHRs) presents a promising strategy to identify patients at heightened risk for readmission and support early intervention.

Aim: This study evaluates the performance of four ML models - logistic regression, random forest, XGBoost, and deep neural networks (DNN) - in predicting 30-day hospital readmissions. It also identifies the most influential predictors using SHapley Additive exPlanations (SHAP) values.

Materials and methods: We conducted a retrospective analysis on 101,766 de-identified inpatient encounters from the University of California, Irvine (UCI) Diabetes 130-United States (US) Hospitals dataset. After preprocessing, including feature imputation, scaling, and one-hot encoding, we trained and validated models on an 80/20 train-test split. Evaluation metrics included accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC).

Results: XGBoost achieved the highest AUC-ROC (0.667), followed closely by logistic regression (0.642) and random forest (0.630). Despite DNNs demonstrating the highest recall for the positive class (0.143), their AUC-ROC (0.579) and precision (0.186) indicated lower reliability. SHAP analysis revealed that previous admissions, number of medications, and comorbidity indicators such as diabetes medication usage and admission type were key predictors influencing model decisions.

Conclusion: XGBoost outperformed other models in predicting 30-day readmissions using EHR data, balancing performance and interpretability when coupled with SHAP values. These findings underscore the promise of ensemble models in improving discharge planning and reducing preventable readmissions. Future research should explore the inclusion of behavioral and social health features to further enhance predictive accuracy.



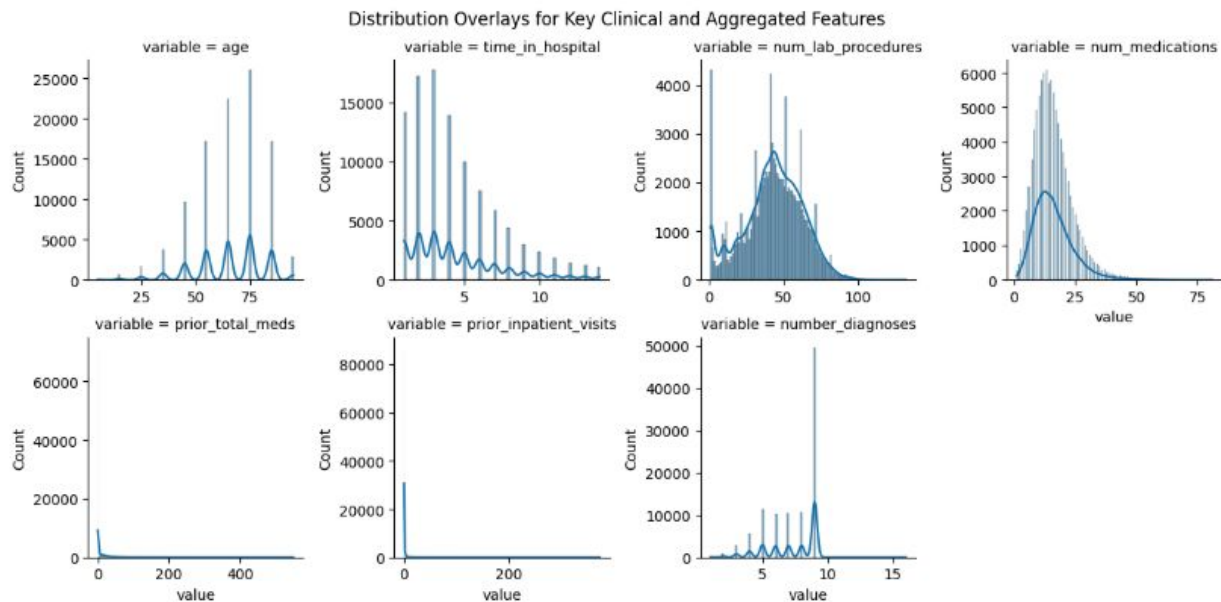
EDA: distribution in key columns

-time_in_hospital, num_lab_procedures, and num_medications show right-skewness and the presence of outliers, which confirm the need to use of StandardScaler to normalize the data.

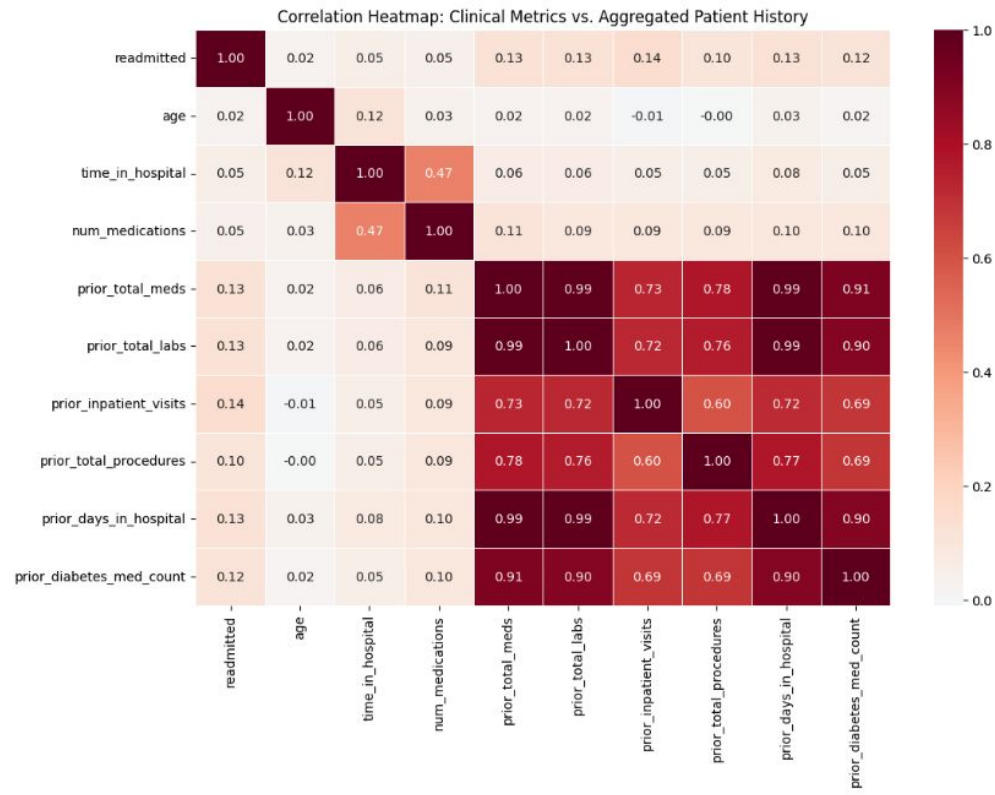
-The age distribution shows various peaks corresponding to the midpoints of original ranges, confirming that the dataset is composed of an elderly population.

-The engineered features prior_total_meds and prior_inpatient_visits show concentrations at zero, showing many patients have no prior hospital history documented within the specific study period.

-number_diagnoses plot has peaks, especially at the value of nine, which might be suggesting some specific administrative practices prevalent during the 1999–2008 timeframe.



-There is also multicollinearity within the aggregated features, with correlations as high as 0.99 between total medications and laboratory procedures. This justifies the selection of ensemble frameworks like XGBoost.





Related work

- Predicting 30-day diabetic readmission is a critical metric for institutional efficiency and healthcare quality in the US.
- Emi-Johnson and Nkrumah (2025) which my study is based on, evaluated four ML models - Logistic Regression, Random Forest, XGBoost, and Deep Neural Networks (DNNs) - using the UCI Diabetes 130-US Hospitals dataset.
- Their findings established XGBoost as the superior performer, reaching a benchmark AUC-ROC of 0.667 over traditional baselines.
- The success of XGBoost (Chen and Guestrin, 2016) stems from its ability to model complex interactions within sparse EHR data, such as medication changes.
- To demonstrate interpretability needed for clinical adoption, the SHAP framework (*the underlying principles of coalitional game theory were established by Lloyd Shapley, 1953; the SHAP framework was introduced by Lundberg and Lee, 2017*) to bring these concepts into machine learning) was applied in the original paper on XGBoost model to evaluate feature contributions.



Methodology

Data Cleaning and Preprocessing

Missing values represented by '?' were converted to NaNs, and features with high missingness, i.e. 'weight' and 'payer_code', along with individual drugs were removed to improve predictive efficiency.

Feature Engineering

Clinical logic was used to group diagnosis codes and specialties, while patient history was captured through cumulative rolling sums to reflect care intensity over multiple visits, avoiding temporal data leakage.

Preprocessing Pipeline

Using an 80/20 train/test split, a ColumnTransformer within a pipeline handled heterogeneous data by applying mode imputation and one-hot encoding for categorical features and StandardScaler for numerical values.

Model Implementation

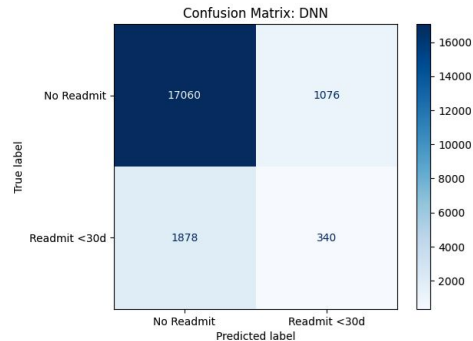
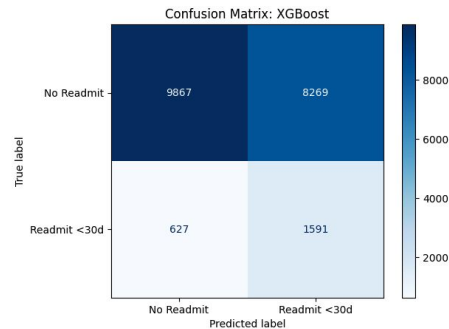
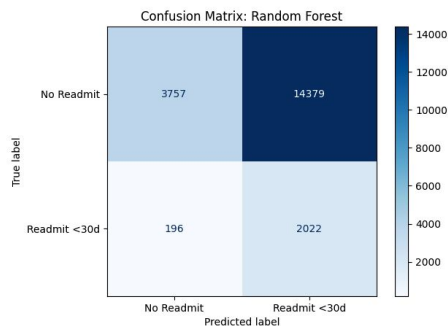
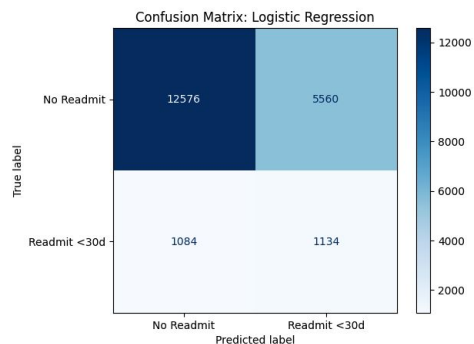
Four machine learning architectures, Logistic Regression, Random Forest, XGBoost, and Deep Neural Networks (DNNs), were developed using scikit-learn pipelines and class weighting to address significant class imbalance in readmission cases.

Evaluation and Optimization

Models were optimized via GridSearchCV/RepeatedStratifiedKFold with three-fold stratified cross-validation focused on AUC-ROC scores; performance was further validated using standard metrics, SHAP interpretability analysis, and the Wilcoxon signed-rank test.

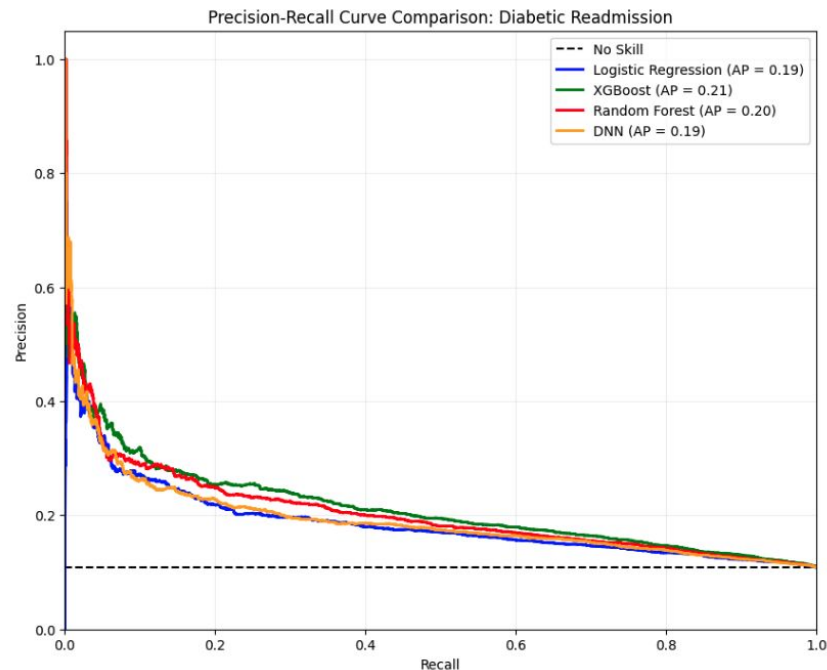
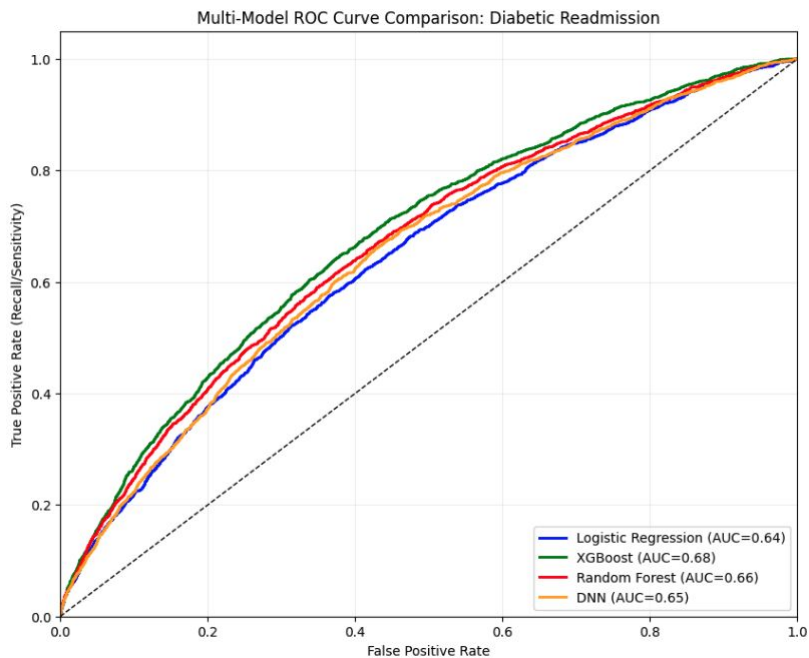


Results: Accuracy, Precision, Recall



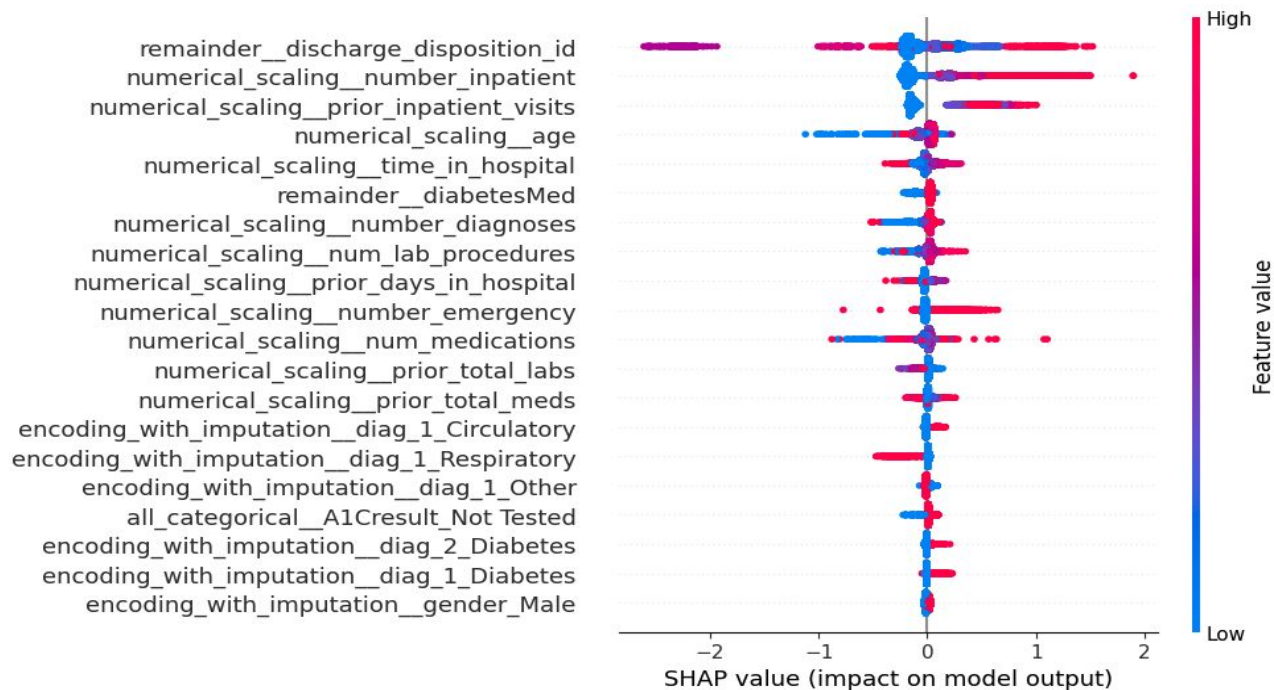
Model	Accuracy	Precision	Recall	AUC-ROC
Logistic Regression	0.6736	0.1694	0.5113	0.6421
Random Forest	0.2839	0.1233	0.9116	0.6644
XGBoost	0.3384	0.1307	0.8972	0.6798
DNNs	0.8549	0.2401	0.1533	0.6509

Results: AUC-ROC and Precision-Recall curves



1) This ROC curve comparison shows that the ensemble models, XGBoost (AUC=0.68) and Random Forest (AUC=0.66), provide better predictive performance and class separability for diabetic readmission compared to the Logistic Regression (AUC=0.64) baseline and the DNN (AUC=0.65). 2) As recall moves toward 1.0 which means catching all readmitted patients, the precision of all models drops toward the baseline. This demonstrates the cost of high sensitivity as to catch every readmitted patient, the hospital must accept a high number of false-positive results - patients who were not actually readmitted although the model flagged as if they would be readmitted.

Results: SHAP analysis for XGBoost



Y axis: the most influential feature is `discharge_disposition_id`, followed by `number_inpatient` and `prior_inpatient_visits`(aggregated column).

X axis: show the impact on the model's output. Points to the right of 0 -> increase the risk of readmission. Points to the left of 0 -> decrease the risk.



Discussion

- The results of this replication demonstrate that the **XGBoost architecture is superior** for predicting 30-day readmissions -> aligning with Emi-Johnson et al. study confirming the model's suitability for high-dimensional tabular EHR data.
- The primary **divergence between the studies lies in the precision-recall trade-off** : my implementation utilised more aggressive class weighting thus prioritising the identification of at-risk patients (recall), leading to a significant drop in accuracy for ensemble methods.
- Clinically, this represents **a shift from a safer model that rarely predicts readmission to a sharper model that predicts almost all 'at-risk' patients but with a trade-off of also misclassifying the ones that are not at risk.**



Limitations

Natural EHR Noise and Cleaning Strategy



Standardised EHR datasets contain inherent variability that limit the achievable predictive accuracy.

The cleaning strategy I used, which is more extensive than in the original study may have simplified the underlying data structure, potentially leading to inflated performance metrics for some models relative to the original noisier dataset.

Missing SDOH Data and Potential Temporal Bias



Unobserved social determinants of health critical for readmission are mostly absent from the dataset -> making prediction less precise.

The feature engineering method I employed, specifically the use of cumulative sums for care intensity, relies on the chronological integrity of the dataset -> if there are any errors in the recording of encounter sequences, a temporal bias might be introduced.

Resource Constraints



High recall model generates a lot of false positives -> this can overwhelm clinical practitioners/hospitals.



Conclusion

- Gradient-boosting (specifically XGBoost) provides superior performance for readmission prediction using structured, high-dimensional EHR data.
- Longitudinal features focusing on rolling care intensity and historical frequency offer significantly higher predictive value than single snapshots.
- High false-positive rates in ensemble models in my study create precision-recall trade-offs that risk overwhelming clinical resources.
- Future iterations must integrate external socioeconomic data and advanced sampling (e.g., SMOTE) to mitigate dataset noise and enhance model precision.



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